1.3 Linear Algebra: Solutions

These are solutions to problems.md. The numerical answers (Problems 4, 5, 6, 15, 16, 17) were checked with a computer algebra system.

Warm-up

1.

(a) **Yes.** This is the null space of the linear map $x \mapsto x_1+2x_2-x_3$, and null spaces are subspaces. You can also check directly: it contains 0 and is closed under addition and scaling, because the condition is linear.
(b) **No.** $(1,0,0)$ and $(0,1,0)$ are in the set, but their sum $(1,1,0)$ is not.
(c) **No.** It does not contain 0.
(d) **Yes.** The zero function satisfies $0 = 2 \cdot 0$. If $f(0) = 2f(1)$ and $g(0) = 2g(1)$, then $(af + bg)(0) = af(0) + bg(0) = 2(af + bg)(1)$. Equivalently, it is the null space of the linear functional $f \mapsto f(0) - 2f(1)$.

2. Since $\dim \mathbb{R}^3 = 3$, it is enough to show the three vectors are independent. Suppose $a(1,1,0)+b(1,0,1)+c(0,1,1) = 0$. Then $a+b = 0$, $a+c = 0$ and $b+c = 0$. Adding all three gives $2(a+b+c) = 0$, so $a+b+c = 0$. Subtracting each equation from this gives $c = 0$, $b = 0$ and $a = 0$. So the list is a basis. (Equivalently, the determinant of the matrix with these columns is $-2 \neq 0$.)

Over $\mathbb{F}_2$ the list is **not** a basis: $(1,1,0)+(1,0,1)+(0,1,1) = (2,2,2) = (0,0,0)$. The step that fails is dividing by 2, which is not possible because $2 = 0$ in $\mathbb{F}_2$. The determinant is $-2 = 0$ in $\mathbb{F}_2$.

3. $D1 = 0$, $Dx = 1$, $Dx^2 = 2x$ and $Dx^3 = 3x^2$. The columns of the matrix are the coordinates of these images:

$$M(D) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

- null D is the constants, $\operatorname{span}(1)$, of dimension 1.
- range $D = \mathcal{P}_2(\mathbb{R})$, of dimension 3. This agrees with rank–nullity: $1+3 = 4$.
- $D^4 = 0$, but $D^3(x^3) = 6 \neq 0$. So the minimal polynomial is z^4.
- The only eigenvalue is 0, and $\dim E(0,D) = 1$, so there is a single Jordan block: $J = J_4(0)$. Concretely, in the basis $6, 6x, 3x^2, x^3$ the matrix of D is exactly $J_4(0)$, since $D(x^3) = 3x^2$, $D(3x^2) = 6x$, $D(6x) = 6$ and $D(6) = 0$.

4. Write $d_n = \det A_n$. Expand along the first row. The $(1,1)$ minor is A_{n-1}. The $(1,2)$ minor has first column $(1,0,\dots,0)^t$; expanding it down that column gives $\det A_{n-2}$. So

$$d_n = 2d_{n-1} - 1 \cdot d_{n-2}, \qquad d_1 = 2,\ d_2 = 3.$$

So $d_3 = 4$ and $d_4 = 5$. By induction, $d_n = n+1$: if $d_{n-1} = n$ and $d_{n-2} = n-1$, then $d_n = 2n-(n-1) = n+1$.

5. $\chi_A(z) = (z-2)^2 - 1 = (z-1)(z-3)$, so the eigenvalues are 1 and 3.

- For $\lambda = 3$: $(A-3I)v = 0$ gives $v = (1,1)$.
- For $\lambda = 1$: $v = (1,-1)$.

With $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, we get $P^{-1}AP = \operatorname{diag}(3,1)$. Then $A^n = P\operatorname{diag}(3^n,1)P^{-1}$, which works out

to

$$A^n = \frac{1}{2}\begin{pmatrix} 3^n+1 & 3^n-1 \\ 3^n-1 & 3^n+1 \end{pmatrix}.$$

Check: $n=1$ gives A. For $n=5$ the formula gives entries 122 and 121, which matches direct multiplication.

6.

- $\|1\|^2 = \int_{-1}^{1} 1\,dx = 2$, so $e_1 = 1/\sqrt{2}$.
- $\langle x, e_1\rangle = \frac{1}{\sqrt{2}}\int_{-1}^{1} x\,dx = 0$, so $f_2 = x$. Then $\|x\|^2 = \int x^2 = 2/3$, so $e_2 = \sqrt{3/2}\,x$.
- $\langle x^2, e_1\rangle = \frac{1}{\sqrt{2}}\cdot\frac{2}{3}$, so the component along e_1 is $\frac{1}{\sqrt{2}}\cdot\frac{2}{3}\cdot\frac{1}{\sqrt{2}} = \frac{1}{3}$. Also $\langle x^2, e_2\rangle = 0$, since x^3 is odd. So $f_3 = x^2 - \frac{1}{3}$, and $\|f_3\|^2 = \int_{-1}^{1}(x^4 - \frac{2}{3}x^2 + \frac{1}{9})dx = \frac{2}{5} - \frac{4}{9} + \frac{2}{9} = \frac{8}{45}$.

The result is

$$e_1 = \tfrac{1}{\sqrt{2}}, \quad e_2 = \sqrt{\tfrac{3}{2}}\,x, \quad e_3 = \sqrt{\tfrac{45}{8}}\,\left(x^2 - \tfrac{1}{3}\right).$$

These are the normalised Legendre polynomials.

Core

7. (a) By rank–nullity, $\dim\operatorname{null} T = \dim V - \dim\operatorname{range} T \geq \dim V - \dim W > 0$. So $\operatorname{null} T \neq \{0\}$, and T is not injective.

(b) $U \times W$ is a vector space under componentwise operations, of dimension $\dim U + \dim W$: if u_i and w_j are bases, then the $(u_i, 0)$ together with the $(0, w_j)$ form a basis. The map $\Gamma(u,w) = u+w$ is linear with range $U+W$. Its null space is $\{(u,-u) : u \in U \cap W\}$, which is isomorphic to $U \cap W$ via $(u,-u) \mapsto u$. Rank–nullity gives $\dim U + \dim W = \dim(U\cap W) + \dim(U+W)$.

8. Every $v \in V$ can be written $v = (v - Pv) + Pv$. Here $Pv \in \operatorname{range} P$, and $P(v-Pv) = Pv - P^2v = 0$, so $v - Pv \in \operatorname{null} P$. Hence $V = \operatorname{null} P + \operatorname{range} P$.

The sum is direct. If $v \in \operatorname{null} P \cap \operatorname{range} P$, write $v = Pw$. Then $0 = Pv = P^2w = Pw = v$.

On range P, P acts as the identity: $P(Pw) = Pw$. On null P it acts as 0. Joining a basis of null P to a basis of range P gives a basis of eigenvectors, so P is diagonalisable with eigenvalues in $\{0,1\}$.

Alternatively: P satisfies $z^2 - z = z(z-1)$, which has distinct roots, so m_P divides it and P is diagonalisable. Both eigenvalues occur unless $P = 0$ or $P = I$.

9. In the standard basis, $T(1,0) = (1,0)$ and $T(0,1) = (1,2)$, so $A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$.

Directly: $T(1,0) = (1,0) = 1\cdot b_1 + 0\cdot b_2$ and $T(1,1) = (2,2) = 0\cdot b_1 + 2\cdot b_2$. So $[T]_{\mathcal{B}'} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$.

By formula: P has the new basis vectors as columns, $P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, so $P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$. Then $AP = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}$ and $P^{-1}AP = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$. The two methods agree. $\mathcal{B}'$ is an eigenbasis.

10. $\psi \in \operatorname{null} T'$ iff $\psi \circ T = 0$ iff $\psi(Tv) = 0$ for all v iff $\psi \in (\operatorname{range} T)^0$.

Then

$$\dim\operatorname{range} T' = \dim W' - \dim\operatorname{null} T' = \dim W - (\dim W - \dim\operatorname{range} T) = \dim\operatorname{range} T,$$

using rank–nullity for T' and $\dim U^0 = \dim W - \dim U$.

Now let $A = M(T)$ with respect to bases of V and W. With respect to the dual bases, $M(T') = A^t$. The column rank of A is $\dim \operatorname{range} T$. The column rank of A^t, which is the row rank of A, is $\dim \operatorname{range} T'$. These are equal.

11. Both matrices are upper triangular with diagonal $2, 2, 3$, so $\chi_A = \chi_B = (z-2)^2(z-3)$.

For B: $(B-2I)(B-3I) = \operatorname{diag}(0,0,1)\operatorname{diag}(-1,-1,0) = 0$. So $m_B = (z-2)(z-3)$, and B is diagonalisable (it is already diagonal).

For A: $A - 2I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $A - 3I = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. The $(1,2)$ entry of the product is $0\cdot 1 + 1\cdot(-1) + 0 = -1 \neq 0$, so $(z-2)(z-3)$ does not annihilate A. Since m_A divides χ_A and has both 2 and 3 as roots, $m_A = (z-2)^2(z-3)$. It has a repeated root, so A is **not** diagonalisable. Also $\dim E(2, A) = 3 - \operatorname{rank}(A - 2I) = 3 - 2 = 1 < 2$.

A and B are **not similar**: similar matrices have the same minimal polynomial. Equivalently, one is diagonalisable and the other is not.

12. T satisfies $p(z) = z^3 - z = z(z-1)(z+1)$, so $m_T \mid p$. Every divisor of p splits over $\mathbb{R}$ with distinct roots, so T is diagonalisable by the diagonalisability theorem. The eigenvalues are roots of m_T, hence of p, so they lie in $\{-1, 0, 1\}$.

13. Let $v_1, \dots, v_m$ be eigenvectors with distinct eigenvalues $\lambda_1, \dots, \lambda_m$. Suppose they are dependent. Let k be the smallest index with $v_k \in \operatorname{span}(v_1, \dots, v_{k-1})$. Such a k exists by the linear dependence lemma, and $k \geq 2$ since $v_1 \neq 0$. Then $v_1, \dots, v_{k-1}$ are independent. Write $v_k = \sum_{i<k} a_i v_i$ and apply $T - \lambda_k I$:

$$0 = \sum_{i<k} a_i(\lambda_i - \lambda_k)v_i.$$

By independence, $a_i(\lambda_i - \lambda_k) = 0$, and $\lambda_i \neq \lambda_k$, so every $a_i = 0$. Then $v_k = 0$, which contradicts the fact that v_k is an eigenvector.

14. Let $Tv = \lambda v$ with $v \neq 0$. Then

$$\lambda\|v\|^2 = \langle Tv, v\rangle = \langle v, Tv\rangle = \langle v, \lambda v\rangle = \bar{\lambda}\|v\|^2,$$

so $\lambda = \bar{\lambda}$. If also $Tw = \mu w$ with $\mu \neq \lambda$ (both real), then $\lambda\langle v, w\rangle = \langle Tv, w\rangle = \langle v, Tw\rangle = \mu\langle v, w\rangle$. Hence $(\lambda - \mu)\langle v, w\rangle = 0$, so $\langle v, w\rangle = 0$.

15. Expand $\det(zI - A)$:

$$\chi_A(z) = (z-2)^3 - 2(z-2) = (z-2)\big((z-2)^2 - 2\big).$$

So the eigenvalues are 2, $2 + \sqrt{2}$ and $2 - \sqrt{2}$.

- $\lambda = 2$: $A - 2I = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$, so $y = 0$ and $x = -z$, giving $v = (1, 0, -1)$.
- $\lambda = 2 + \sqrt{2}$: the rows of $A - \lambda I$ are $(-\sqrt{2}, -1, 0)$, $(-1, -\sqrt{2}, -1)$ and $(0, -1, -\sqrt{2})$. The first gives $y = -\sqrt{2}x$ and the third gives $z = -y/\sqrt{2} = x$. So $v = (1, -\sqrt{2}, 1)$.
- $\lambda = 2 - \sqrt{2}$: similarly $v = (1, \sqrt{2}, 1)$.

The three eigenvectors are mutually orthogonal, as Problem 14 predicts. Their norms are $\sqrt{2}$, 2 and 2. So

$$Q = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \qquad D = \operatorname{diag}(2,\, 2+\sqrt{2},\, 2-\sqrt{2}),$$

and $A = QDQ^t$.

16. Solve $(A - \lambda I)v = 0$ for each eigenvalue:

- $\lambda = 1$: $v_1 = (-1, 1, 0, 0)$.
- $\lambda = 2$: $v_2 = (1, -1, 0, 1)$.
- $\lambda = 4$: $\operatorname{rank}(A - 4I) = 3$, so $E(4, A)$ is 1-dimensional, spanned by $u = (1, 0, -1, 1)$.

The algebraic multiplicity of 4 is 2, but its geometric multiplicity is 1. So there is one Jordan block $J_2(4)$ for $\lambda = 4$, and

$$J = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 4 \end{pmatrix}.$$

For the Jordan chain, we need w with $(A - 4I)w = u$. The first column of $A - 4I$ is $(5 - 4, 0, -1, 1) = (1, 0, -1, 1) = u$, so $w = e_1 = (1, 0, 0, 0)$ works. Take

$$P = \begin{pmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

with columns v_1, v_2, u, w. Then $Au = 4u$ and $Aw = u + 4w$, so the last two columns produce the block $J_2(4)$, and $P^{-1}AP = J$. You can check $AP = PJ$ column by column.

17. (a) $\dim V' = \dim V = 3$, so it suffices to show independence. Suppose $a\mathrm{ev}_0 + b\mathrm{ev}_1 + c\mathrm{ev}_2 = 0$. Apply it to the polynomials p_0, p_1, p_2 from part (b). Each p_j is 1 at j and 0 at the other two points, so this gives $a = 0$, $b = 0$ and $c = 0$ in turn.

(b) We need $\mathrm{ev}_i(p_j) = p_j(i) = \delta_{ij}$. These are the Lagrange interpolation polynomials:

$$p_0 = \tfrac{(x-1)(x-2)}{2} = \tfrac{1}{2}x^2 - \tfrac{3}{2}x + 1, \quad p_1 = \tfrac{x(x-2)}{-1} = -x^2 + 2x, \quad p_2 = \tfrac{x(x-1)}{2} = \tfrac{1}{2}x^2 - \tfrac{1}{2}x.$$

They form a basis because they are 3 vectors in a 3-dimensional space and are independent: apply ev_i to $\sum c_j p_j = 0$ to get $c_i = 0$.

(c) Any $p \in V$ satisfies $p = \sum_i p(i)\, p_i$, because both sides take the same value at $0, 1, 2$ and a nonzero polynomial of degree ≤ 2 has at most 2 roots. So $p = p_0 + 3p_1 + 7p_2 = x^2 + x + 1$. Check: $p(0) = 1$, $p(1) = 3$, $p(2) = 7$.

18. (a) The quotient map $\pi : V \to V/U$ is linear and surjective with $\operatorname{null}\pi = U$. Rank–nullity gives $\dim V = \dim U + \dim V/U$.

(b) Let $N = \operatorname{null} T$ and define $\tilde{T}(v + N) = Tv$.

- *Well defined:* if $v + N = v' + N$, then $v - v' \in N$, so $Tv - Tv' = T(v - v') = 0$.
- *Linear:* this follows from linearity of T and the definition of the operations on V/N.
- *Injective:* $\tilde{T}(v + N) = 0$ means $Tv = 0$, so $v \in N$ and $v + N = 0 + N$.
- *Surjective onto* $\operatorname{range} T$*:* this is clear from the definition.

(c) Define $\bar{T}(v+U) = Tv$. If $v - v' \in U \subseteq \operatorname{null} T$, then $Tv = Tv'$, so $\bar{T}$ is well defined. Linearity is as in (b). This is the *universal property of the quotient*: $T = \bar{T} \circ \pi$.

19. Let $A = \begin{pmatrix} a & b \\ c & d' \end{pmatrix}$, so $t = a + d'$ and $d = ad' - bc$. Then

$$A^2 = \begin{pmatrix} a^2 + bc & ab + bd' \\ ca + d'c & bc + d'^2 \end{pmatrix}, \quad tA = \begin{pmatrix} a^2 + ad' & ab + bd' \\ ac + cd' & ad' + d'^2 \end{pmatrix}.$$

So $A^2 - tA = \begin{pmatrix} bc - ad' & 0 \\ 0 & bc - ad' \end{pmatrix} = -dI$, and $A^2 - tA + dI = 0$.

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$: $t = 5$ and $d = -2$, so $A^2 - 5A - 2I = 0$. Rearranging, $A(A - 5I) = 2I$, so

$$A^{-1} = \tfrac{1}{2}(A - 5I) = \tfrac{1}{2}\begin{pmatrix} -4 & 2 \\ 3 & -1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}.$$

Check: $AA^{-1} = \begin{pmatrix} -2+3 & 1-1 \\ -6+6 & 3-2 \end{pmatrix} = I$.

Challenge

20. Let $\dim V = n \geq 1$ and choose $v \neq 0$. The list $v, Tv, \dots, T^n v$ has $n+1$ vectors, so it is dependent: there are $a_0, \dots, a_n$, not all zero, with $\sum a_j T^j v = 0$. Let $p(z) = \sum a_j z^j \neq 0$. Then p is not a nonzero constant, because that would give $a_0 v = 0$ with $a_0 \neq 0$. So $\deg p = m \geq 1$.

This is where $\mathbb{C}$ is used. By the Fundamental Theorem of Algebra, $p(z) = c(z - \lambda_1) \cdots (z - \lambda_m)$ with $c \neq 0$. Hence

$$0 = p(T)v = c(T - \lambda_1 I) \cdots (T - \lambda_m I)v.$$

If every $T - \lambda_j I$ were injective, the composition would be injective, and $p(T)v \neq 0$. So some $T - \lambda_j I$ has a nonzero null vector, and λ_j is an eigenvalue.

Over $\mathbb{R}$, p might have no real roots, and the argument breaks. Rotation by $\pi/2$ shows that the conclusion really does fail.

21. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of S, so $V = \bigoplus_j E(\lambda_j, S)$.

Each $E_j = E(\lambda_j, S)$ is T-invariant: if $Sv = \lambda_j v$, then $S(Tv) = TSv = \lambda_j Tv$.

$T|_{E_j}$ is diagonalisable: $m_T(T|_{E_j}) = m_T(T)|_{E_j} = 0$, so the minimal polynomial of $T|_{E_j}$ divides m_T. That polynomial splits with distinct roots, hence so does every divisor.

So each E_j has a basis of eigenvectors of $T|_{E_j}$. Each of these vectors is also an eigenvector of S, with eigenvalue λ_j. Joining these bases over j gives a basis of V of common eigenvectors.

22. Suppose T is upper triangular with respect to the basis $v_1, \dots, v_n$. Equivalently, $Tv_k \in \operatorname{span}(v_1, \dots, v_k)$ for each k, i.e. each $U_k = \operatorname{span}(v_1, \dots, v_k)$ is T-invariant.

Apply Gram–Schmidt to $v_1, \dots, v_n$ to get an orthonormal basis $e_1, \dots, e_n$ with $\operatorname{span}(e_1, \dots, e_k) = U_k$ for every k. Then $Te_k \in T(U_k) \subseteq U_k = \operatorname{span}(e_1, \dots, e_k)$. So the matrix of T with respect to $e_1, \dots, e_n$ is upper triangular.

(The assumption holds over $\mathbb{C}$. By induction using Problem 20: take an eigenvector v_1, then apply the induction hypothesis to the operator that T induces on $V/\operatorname{span}(v_1)$.)

23. (a) Write $N = \bigoplus_i J_{k_i}(0)$. A Jordan block $J_k(0)$ has $J_k(0)^{k-1} \neq 0$ and $J_k(0)^k = 0$. So $N^r = 0$ iff $r \geq \max k_i$, and $m_N = z^{\max k_i}$. Each block has a 1-dimensional null space, spanned by its first basis vector, so $\dim \operatorname{null} N$ is the number of blocks.

(b) Take

$$N_1 = J_3(0) \oplus J_3(0) \oplus J_1(0), \qquad N_2 = J_3(0) \oplus J_2(0) \oplus J_2(0).$$

Both are 7×7 with minimal polynomial z^3 and nullity 3, since each has 3 blocks.

They are not similar. If $N_2 = PN_1P^{-1}$, then $N_2^2 = PN_1^2P^{-1}$, so $\operatorname{rank} N_1^2 = \operatorname{rank} N_2^2$. But $J_3(0)^2$ has rank 1 and $J_2(0)^2 = J_1(0)^2 = 0$. So $\operatorname{rank} N_1^2 = 1+1+0 = 2$, while $\operatorname{rank} N_2^2 = 1+0+0 = 1$. Contradiction.

In general, the ranks of all the powers N^k determine the partition. Minimal polynomial and nullity record only the largest part and the number of parts.

24. (a) $\operatorname{tr}(AB) = \sum_i \sum_j A_{ij}B_{ji} = \sum_j \sum_i B_{ji}A_{ij} = \operatorname{tr}(BA)$. So $\operatorname{tr}(AB - BA) = 0$. But $\operatorname{tr} I_n = n \neq 0$ in $\mathbb{R}$, so $AB - BA = I$ is impossible. The same argument works over any field in which $n \neq 0$, in particular any field of characteristic 0.

(b) *Well defined:* X maps the ideal (x^p) into itself. For D, $D(x^p g) = px^{p-1}g + x^p g' = x^p g'$, because $p = 0$ in $\mathbb{F}_p$. So D also maps (x^p) into itself, and both induce linear maps on the quotient V.

The commutator: for any f, the product rule gives

$$(DX - XD)f = D(xf) - xDf = f + xf' - xf' = f.$$

So $DX - XD = I$ on V, which has dimension p.

No contradiction: $\operatorname{tr} I_p = p = 0$ in $\mathbb{F}_p$, so the trace argument in (a) gives no obstruction. The argument in (a) needs $n \neq 0$ in the field.